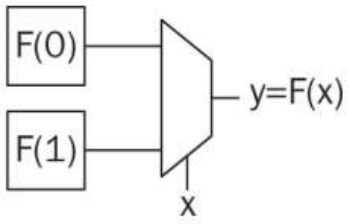
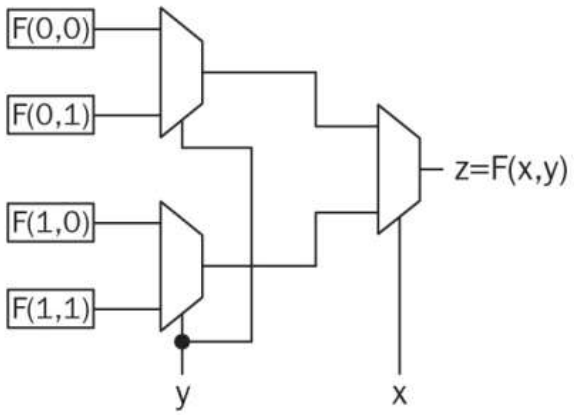
WEEK7 LAB ASSIGNMENT

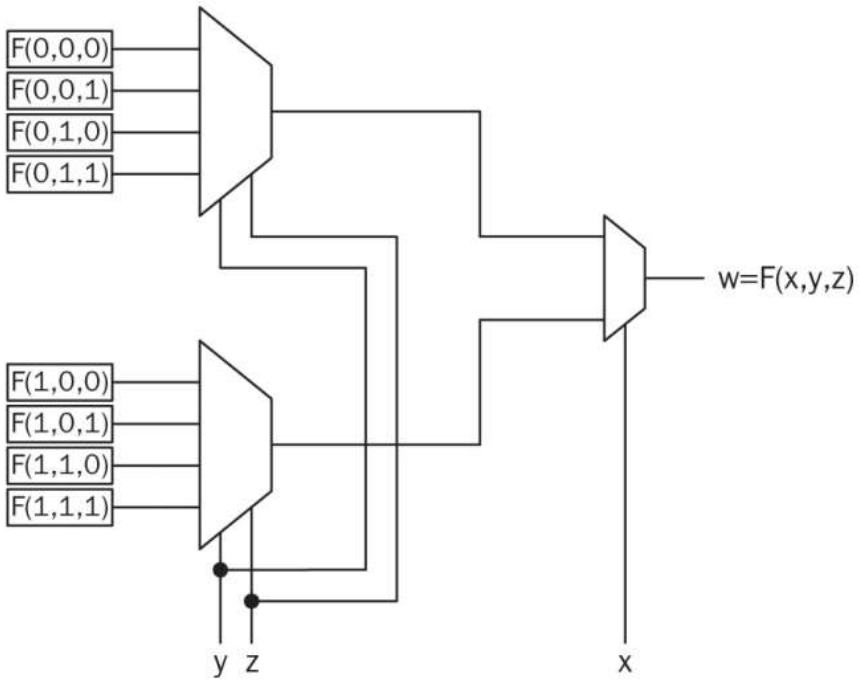
a) Writa a module for a 2:1 mux



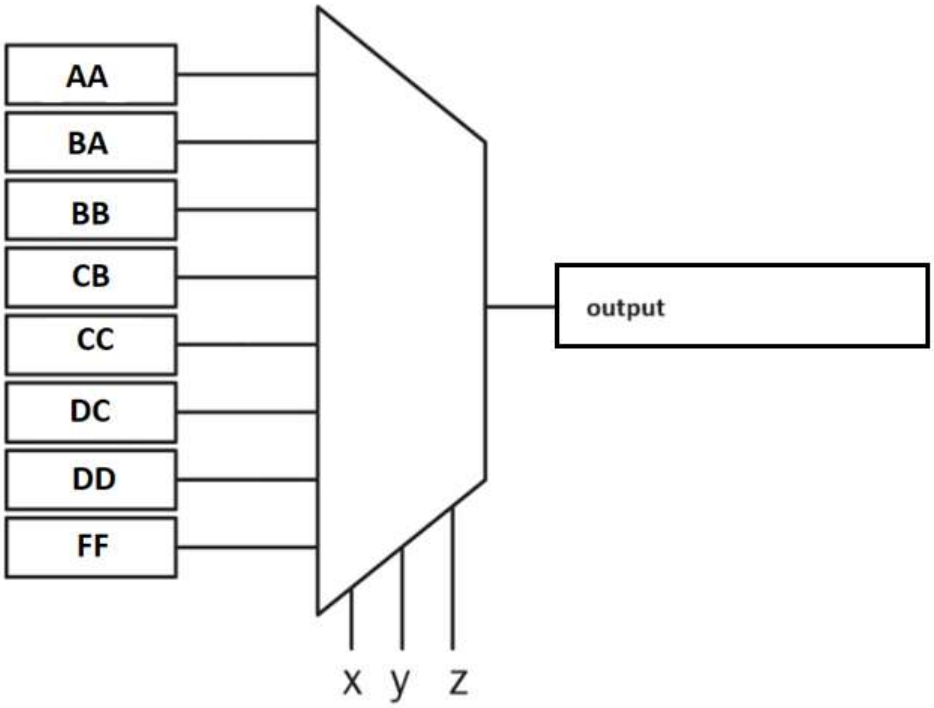
b) Compose another module using (a) to have a 4:1 mux



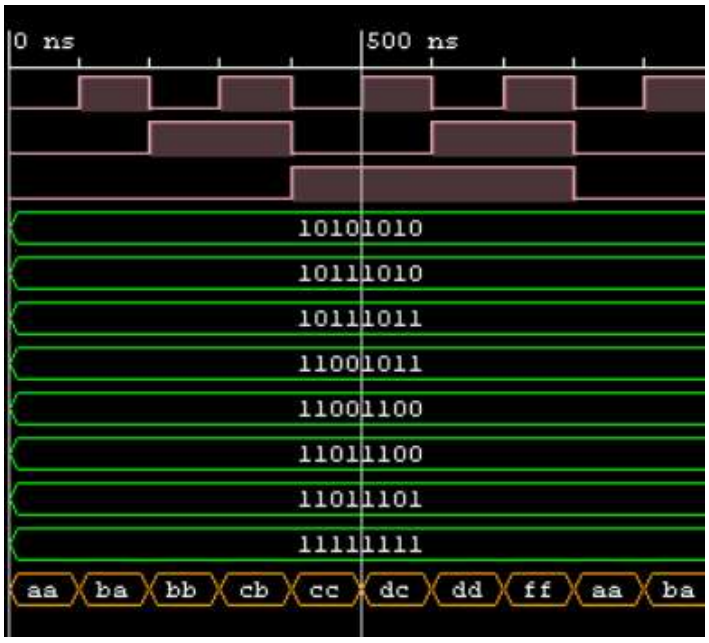
c) Compose another module using (a) and (b) to have a 8:1 mux



d) Now, write your main module which includes a LUT that contains eight hexadecimal constants: aa,ba,bb,cb,cc,dc,dd,ff



e) Simulation part: In your testbench file, using “repeat” and an appropriate time delay show each item of look up table in order, as shown in below figure.



HINT:

i) Use mux 2 to 1 code below for (a) part.

```
 3 module mux2to1(x, F0, F1, y);
 4
 5 output  [7:0] y;
 6 input [7:0]  F0, F1;
 7 input x;
 8
 9 assign y =(x)?F1:F0;
10
11 endmodule
```

ii) Use hierarchical module representation (5.1.3 in the book) for parts (b) and (c).

iii) RTL schematic of the top module will be similar to one below.

